

Bradley Eugene Sakran

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EDUCATION

University of British Columbia

Bachelor of Science in Computer Science (GPA 3.40)

Sep. 2023 – May. 2028

Vancouver, Canada

WORK EXPERIENCE

Research Assistant

University of British Columbia

Sep. 2025 – Present

Vancouver, Canada

- Built an agentic survey platform with **FastAPI**, integrating **Model Context Protocol (MCP)** servers with namespaced tool routing to retrieve demographic-specific political party support data.
- Designed a **PostgreSQL** identity caching layer to persist MCP tool results per respondent.
- Engineered dynamic system prompt construction with **Pydantic**-validated demographic filters, personalizing LLM behavior per Qualtrics respondent across persistent conversation sessions.
- Built an asynchronous CSV ingestion pipeline with **Celery** and **Redis**, using **Pandas** to replace party support datasets, exposing ingestion progress via a polling endpoint.

AI Engineer Co-op

University of British Columbia

May. 2025 – Aug. 2025

Vancouver, Canada

- Built a modular Retrieval-Augmented Generation (RAG) pipeline with **Python (FastAPI)**.
- Engineered an asynchronous PDF ingestion pipeline using **PyMuPDF** to extract texts, tables, and images, dispatched via **Redis** message queues to **Celery** workers for processing.
- Built an async session cache layer with **Redis** for real-time conversation state and **PostgreSQL** for long-term persistence, with **Celery** workers triggering summarization on session close.

Machine Learning Engineer Intern

OLX

Jun. 2024 – Aug. 2024

Jakarta, Indonesia

- Built a retraining pipeline on **AWS EC2** using **Python**, **TensorFlow**, Keras, and ClearML, automating training/deployment of **Bi-LSTM** (intent detection) and **Word2Vec** models.
- Implemented a semi-supervised labeling algorithm for **28.45 MB** of unlabeled chat records using vectorization, **DBSCAN** clustering, and topic modeling, improving training efficiency.
- Optimized preprocessing of large-scale datasets from **AWS RDS (PostgreSQL)** using pandas, NumPy, scikit-learn, and parallelization with pandarallel, including text normalization (**stemming**).
- Achieved **355.6 KBps** throughput on low-tier EC2 by streamlining preprocessing and applying concurrent hyperparameter tuning with **Optuna**, raising model accuracy by **26%**.
- Integrated Deepchecks for continuous data validation and model registry on **AWS S3**; followed Scrum Agile cycles, completing **~8** story points per one-week sprint.

NOTABLE PROJECT

b.env [nwHacks '25] – *Go, RabbitMQ, Next.js, Solidity* – GitHub 

Jan. 2025

- Won **1Password: Best Security Hack** and competed among **734** hackers and **182** projects.
- Developed middleware routing requests to third-party services through API keys stored on-chain.
- Connected users via MetaMask, enabling interactions between a Solidity smart contract and an Ethereum wallet.
- Implemented an API queue system with **RabbitMQ** as the message broker, handling high-concurrency requests using **Go (Fiber)** and a **Next.js** web server.

TECHNICAL SKILLS

Languages: Python, Go, JavaScript, TypeScript, Java, C#, C++, C, SQL, HTML/CSS, R, Racket

Frameworks: FastAPI, Gin, Fiber, Django, TensorFlow, LangChain, Node.js, Express, React, Tailwind

Tools: RabbitMQ, Redis, Kafka, ClearML, Git, Docker, AWS, Atlassian, VS Code

Data: PostgreSQL, MongoDB, Pinecone, ChromaDB, Qdrant

